

M17 Review / Repair - Teaching RC2

OPEN ONLY AFTER SAVED ATTEMPT

P01 repair - From requirements to architecture

Architecture diagrams without requirements are decoration; tradeoffs only make sense against a workload.

Repair principle: Collect sources/formats/volumes/change rate and data growth.

Evidence focus: operational -> small team

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P02 repair - Batch vs streaming decisions

Streaming adds operational/state complexity; batch may miss required event latency.

Repair principle: Start from business latency: seconds/minutes/hours/daily.

Evidence focus: historical rebuild -> batch/backfill even in stream system

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P03 repair - Warehouse vs lakehouse

Both can solve analytics; architecture depends on workload and operating model.

Repair principle: Warehouse: SQL-centric managed relational analytics/concurrency/governance.

Evidence focus: both -> hybrid with clear ownership

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P04 repair - Managed service vs build yourself

Building infrastructure yourself gives control but creates 24/7 operational responsibility.

Repair principle: Managed service reduces patching/scaling/HA burden but may cost more per unit and create service limits/vendor coupling.

Evidence focus: cost -> include people + failure cost

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P05 repair - Scaling and bottleneck reasoning

Systems scale until one resource/dependency limits throughput/latency. Randomly adding nodes increases cost without fixing the bottleneck.

Repair principle: Measure source rate, queue/lag, task duration, CPU/memory/I/O/network, shuffle, DB query/load throughput and file counts.

Evidence focus: Kafka hot partition lag -> key/partition skew

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P06 repair - Reliability and failure domains

A reliable architecture expects components/zones/jobs to fail and limits the impact.

Repair principle: Failure domain is a set that can fail together: process, node, zone, region, credential, shared DB, source.

Evidence focus: idempotency -> safe retry/replay

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P07 repair - Cost-aware architecture

Cloud/distributed data systems can scale cost quickly through compute hours, scans, data transfer, duplicated storage and operational people time.

Repair principle: Identify fixed vs variable drivers: capacity/cluster runtime, scanned bytes, storage, egress, API calls, orchestration runs, operational labor.

Evidence focus: people/on-call -> managed service tradeoff

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P08 repair - Security and governance by design

Retrofitting security after all data is copied broadly is difficult and risky.

Repair principle: Classify data/sensitivity and minimize raw PII spread.

Evidence focus: audit -> access/run logs

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.